# Extended Case 3 — BLK Retail Display

From a sales difference to a business decision · 307312 Applied Statistics — Masters Level

- **Start after:** the independent-groups comparison in Part 3.
- **Suggested time:** 50–60 minutes.
- **Role:** apply independent-group comparisons to COLA data and prepare a commercial briefing.

**Learning outcomes:**

- Interpret an independent-groups effect and confidence interval in cases per store per week.
- Separate random assignment from representative sampling.
- Evaluate business importance with explicitly assumed margin and display costs.

## 1. The merchandising manager's brief

- BLK cola is sold in normal-shelf and end-of-aisle displays.
- **Source:** *Basic Business Statistics*, ninth edition, Chapter 10, pp. 336–339.
- **Design:** 20 selected stores of similar sales volume; random assignment of ten stores to each condition.
- **Unit:** one store observed for one week; outcome is cases sold per week.
- **Limit:** random selection from all supermarkets is not established, and one week does not establish long-term effects.
- **Business question:** Does the sales gain warrant further consideration of the display space?

## 2. Reuse the course's canonical COLA file

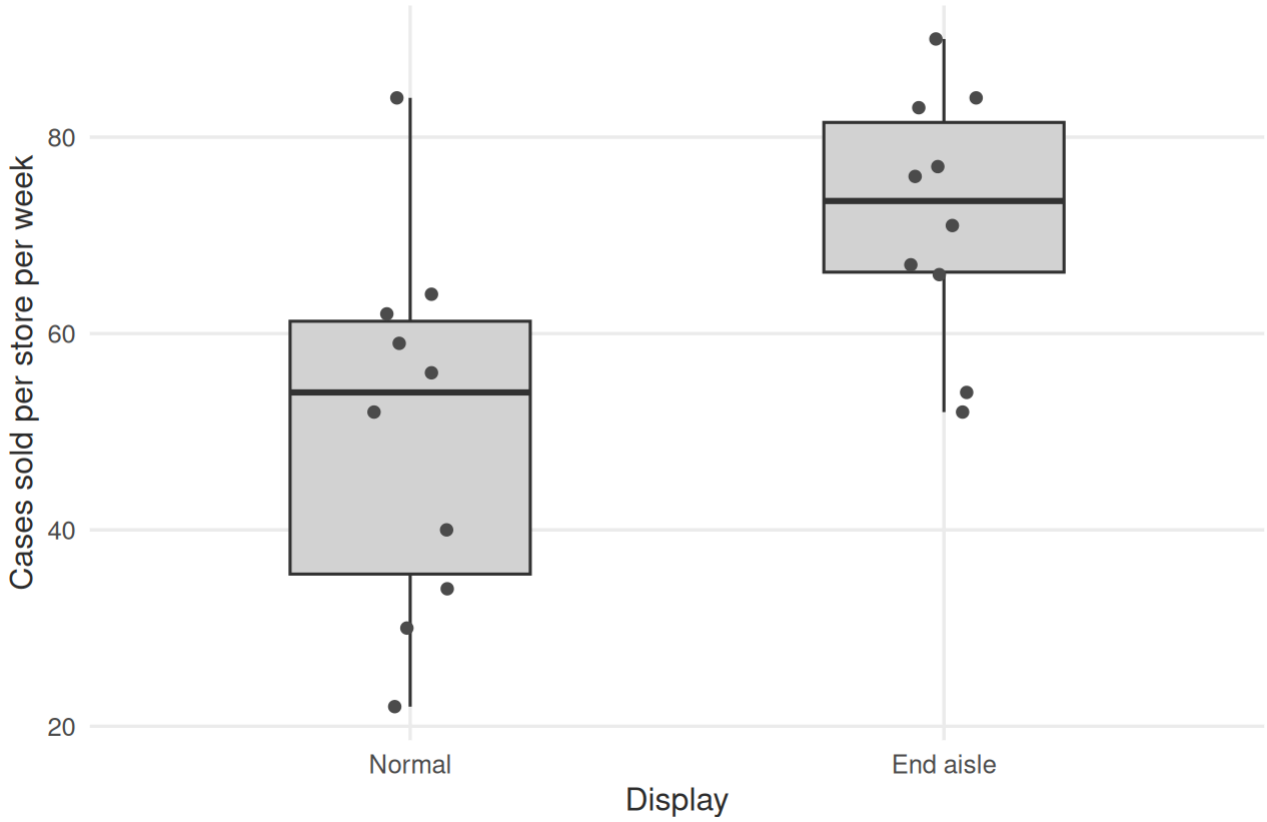
```
source("lessons/_includes/course-tools.R")
cola <- read.csv("data/business/cola.csv")
cola$display <- factor(cola$display, levels = c("Normal", "End aisle"))
normal <- cola$weekly_cases[cola$display == "Normal"]
end <- cola$weekly_cases[cola$display == "End aisle"]
stopifnot(length(normal) == 10, length(end) == 10)
knitr::kable(cola %>% group_by(display) %>%
  summarise(stores = n(), mean_cases = mean(weekly_cases),
    sd_cases = sd(weekly_cases), .groups = "drop"), digits = 2)
```

| display | stores | mean_cases | sd_cases |
|---|---|---|---|
| Normal | 10 | 50.3 | 18.73 |
| End aisle | 10 | 72.0 | 12.54 |

- `data/business/cola.csv`; guide: `data/business/README.md`.
- Each row is one store; the source's side-by-side columns did not represent pairs.
- Course store IDs identify rows; they are not publisher store names.

## 3. Review the worked comparison

```
ggplot(cola, aes(display, weekly_cases)) +
  geom_boxplot(fill = petra_accent, width = .45, outlier.shape = NA) +
  geom_point(position = position_jitter(width = .07, height = 0, seed = 307312),
             colour = petra_ink) +
  labs(x = "Display", y = "Cases sold per store per week")
```



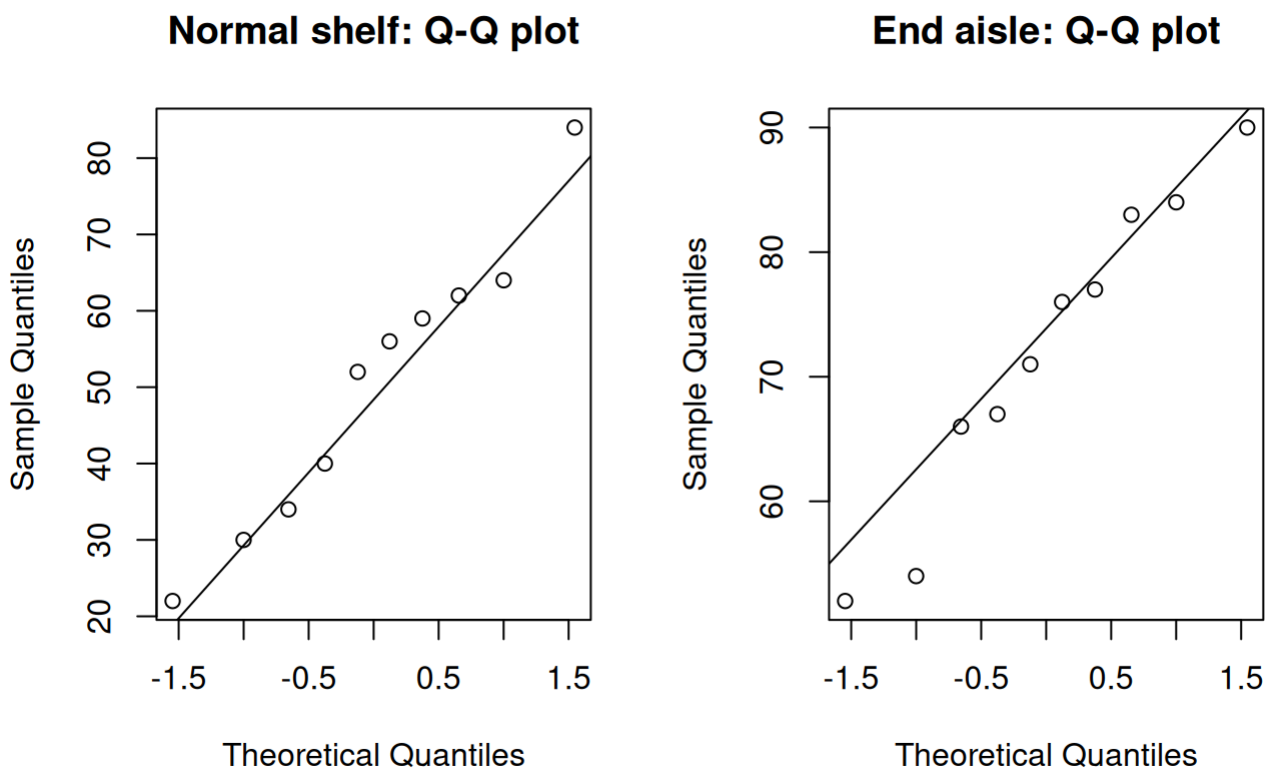

```
welch <- t.test(end, normal, var.equal = FALSE)
knitr::kable(data.frame(difference = mean(end) - mean(normal),
  lower_95 = welch$conf.int[1], upper_95 = welch$conf.int[2],
  p_value = welch$p.value), digits = 4)
```

| difference | lower_95 | upper_95 | p_value |
|---|---|---|---|
| 21.7 | 6.5687 | 36.8313 | 0.0078 |

- The estimated difference is **end aisle minus normal**: 21.7 cases per store per week.
- The Welch 95% interval is about **6.57–36.83 cases**, with p about **0.00785**.
- The two-sided null is equal mean sales under the two conditions.
- The book uses a pooled-variance t test; the course uses Welch, which avoids the equal-variance assumption.
- Random assignment supports the experimental contrast among these stores, subject to implementation and no interference.
- Sales evidence alone does not establish the display's net commercial value.

## 4. Review assumptions before reporting

```
par(mfrow = c(1, 2))
qqnorm(normal, main = "Normal shelf: Q-Q plot"); qqline(normal)
qqnorm(end, main = "End aisle: Q-Q plot"); qqline(end)
```




```
par(mfrow = c(1, 1))
```

- Each group contains only ten stores; Q-Q plots have limited power to reveal departures from normality.
- Independence follows from the design and store operations, not from a normality plot.
- Check whether stores could affect one another's sales or whether promotions differed.
- Do not choose a paired test because the original spreadsheet aligned two columns.

## 5. Independent commercial briefing

1. Reproduce the effect, confidence interval and p-value, with units and direction.
2. Run the textbook's pooled test using `var.equal = TRUE`; compare its interval and assumptions with Welch's.
3. Run a Wilcoxon rank-sum comparison. Explain why it is not generally a test of equality of means.
4. Define the population to which you can reasonably discuss the results; distinguish assignment from selection.
5. Complete the commercial sensitivity analysis below and recommend a next step.

## Hypothetical commercial assumptions

- Contribution margin per additional case: **$4, $6 or $8**.
- Opportunity cost of the display: **$80 per store per week**.

- These are assignment assumptions, not COLA observations or textbook financial data.
- Assume the measured sales difference represents incremental cases and the chosen margin applies to them; discuss why this may fail.

```
margin_per_case <- c(4, 6, 8)
display_cost <- 80
# For each assumed margin, calculate:
# estimated net contribution = estimated case difference * margin - cost
# corresponding interval endpoints = Welch endpoints * margin - cost
# break-even additional cases = cost / margin
```

- Multiplying the sales interval by a fixed positive margin gives a conditional interval under that assumed margin.
- It does not include uncertainty in costs, margins, cannibalization or future display effects.
- Treat a zero-crossing interval as uncertainty about the assumed commercial threshold, not proof of no value.

## Submission

- One results table comparing Welch, pooled and rank-based analyses with their different targets.
- One commercial sensitivity table and one labelled chart.
- A 150–200-word recommendation addressing magnitude, uncertainty, cost assumptions and transfer to other stores or weeks.